

HashVote

A Blockchain-based e-voting system

Group 22

Team Details

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Problem Statement

Elections are a fundamental pillar of a democratic system enabling the general public to express their views in the form of a vote. Due to their significance to our society, the election process should be transparent and reliable so as to ensure participants of its credibility. Within this context, the approach to voting has been an ever-evolving domain. This evolution is primarily driven by the efforts to make the system secure, verifiable and transparent. In view of its significance, continuous efforts have been made to improve overall efficiency and resilience of the voting system. Electronic voting or e-voting has a profound role in this. Since its first use as punched-card ballots in the 1960's, e-voting systems have achieved remarkable progress with its adoption using the internet technologies. However, e-voting systems must adhere to specific benchmark parameters so as to facilitate its widespread adoption. These parameters include anonymity of the voter, integrity of the vote and non-repudiation among others.

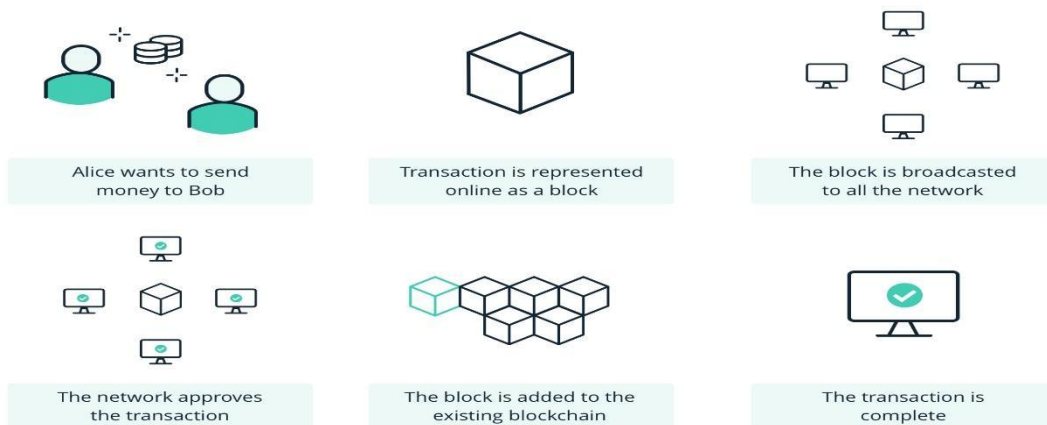
Election processes can be rigged via ballot falsification, coercion, and other such malpractices. A centralized authority, having complete power over vote casting and counting, can easily manipulate the results. Hackers can get into the system even before the process starts and keep rigging the entire process. Furthermore, a person must wait in a long line to cast a vote, which is a time-consuming operation.

What is Blockchain?

Blockchain is one of the emerging technologies with strong cryptographic foundations enabling applications to leverage these abilities to achieve resilient security solutions. A Blockchain resembles a data structure which maintains and shares all the transactions being executed through its genesis. It is primarily a distributed decentralized database that maintains a complete list of constantly germinating and growing data records secured from unauthorized manipulating, tampering and revision.

Blockchain allows every user to connect to the network, send new transactions to it, verify transactions and create new blocks. Each block is assigned a cryptographic hash (which may also be treated as a fingerprint of the block) that remains valid as long as the data in the block is not altered. If any changes are made in the block, the cryptographic hash would change immediately indicating the change in the data which may be due to a malicious activity. Therefore, due to its strong foundations in cryptography, blockchain has been increasingly used to mitigate against unauthorized transactions across various domains

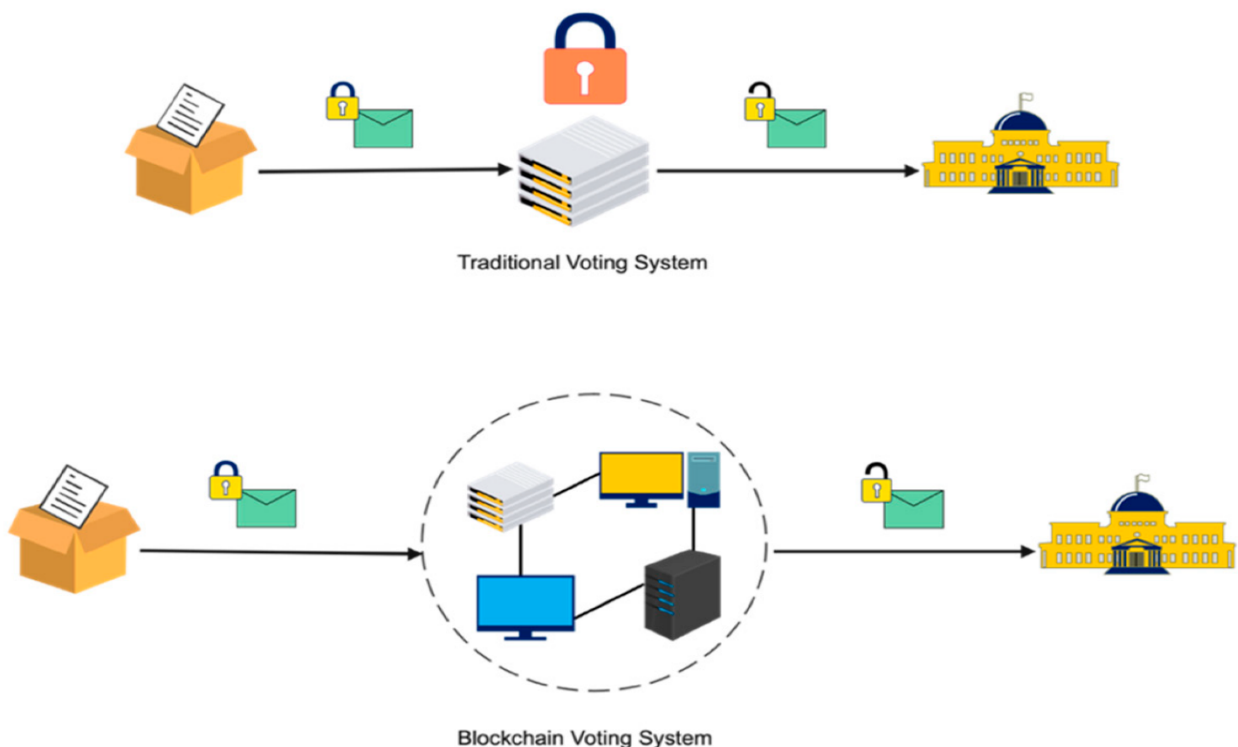
How does blockchain work



Blockchain and E-voting

Blockchain technology fixed shortcomings in today's method in elections, made the polling mechanism clear and accessible, stopped illegal voting, strengthened the data protection, and checked the outcome of the polling.

Today, there is a viable solution to overcome these risks of voting, which is blockchain technology. In the below figure, one can see the main difference between both of the systems. In traditional voting systems, we have a central authority to cast a vote. If someone wants to modify or change the record, they can do it quickly, no one knows how to verify that record. One should not have the central authority, while in blockchain the data are stored in multiple nodes. It is not possible to hack all nodes and change the data. Thus, in this way, one cannot destroy the votes and efficiently verify the votes by tally with other nodes



Zero-Knowledge-Proof

In cryptography, a zero-knowledge proof or zero-knowledge protocol is a method by which one party (the prover) can prove to another party (the verifier) that a given statement is true while the prover avoids conveying any additional information apart from the fact that the statement is indeed true. The essence of zero-knowledge proofs is that it is trivial to prove that one possesses knowledge of certain information by simply revealing it; the challenge is to prove such possession without revealing the information itself or any additional information. We have used the technique of discrete-log to implement ZKP.

In the discrete-log protocol above, the prover first picks a random r and then the verifier picks a random b . It's important that the prover pick first and bit-commit his choice by sending h to the verifier. Otherwise if he saw V 's choice of b he could cheat in one of the two ways given earlier. To make this non-interactive, we have the signer simulate both prover and verifier, and publish the transcript of their whole dialogue. The signer first does a random choice of r for the prover as before, but the verifier's random choice is simulated by hashing the input data and a value computed from the prover's choice of r . By making the verifier's choice depend on the prover's choice, we make it hard for the signer to fake the outcome. The prover must choose r first, then compute h , before he finds out what the verifier's choice for b will be.

In our project, while a user casts a vote, we ask him to verify his secret voter ID using a Zero Knowledge Proof. If the verification is successful, the user is then able to cast the vote.

Code Overview

Client -

The frontend is made using **Bootstrap** framework and **React** library. All the components lie in the components folder, while the base component 'App.js' lies in the root folder. The React components used are listed below :

App.js	Root component, uses react router to define all the routes
Login.js	Verifies the college ID and password of the user
Register.js	Asks for name, college ID and a password to register
Home.js	Lists all the current live polls
Navbar.js	Contains links to create poll, view transactions and to view the blockchain
Card.js	Displays the current polls' details in a bootstrap card
Poll.js	Displays all candidates, a user can cast votes and add candidates
NavigatePoll.js	A navbar with extra option of count votes
VoteCount.js	Counts and lists the votes secured by each candidate
Transactions.js	Lists all the successful transactions done by the user
Blockchain.js	Displays the entire blockchain, can be accessed only with admin access

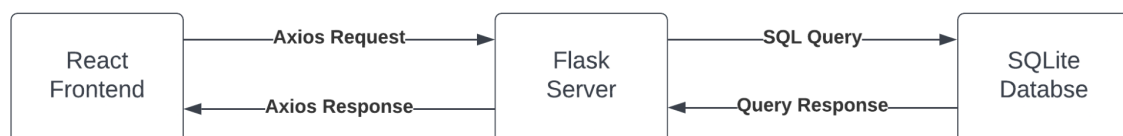
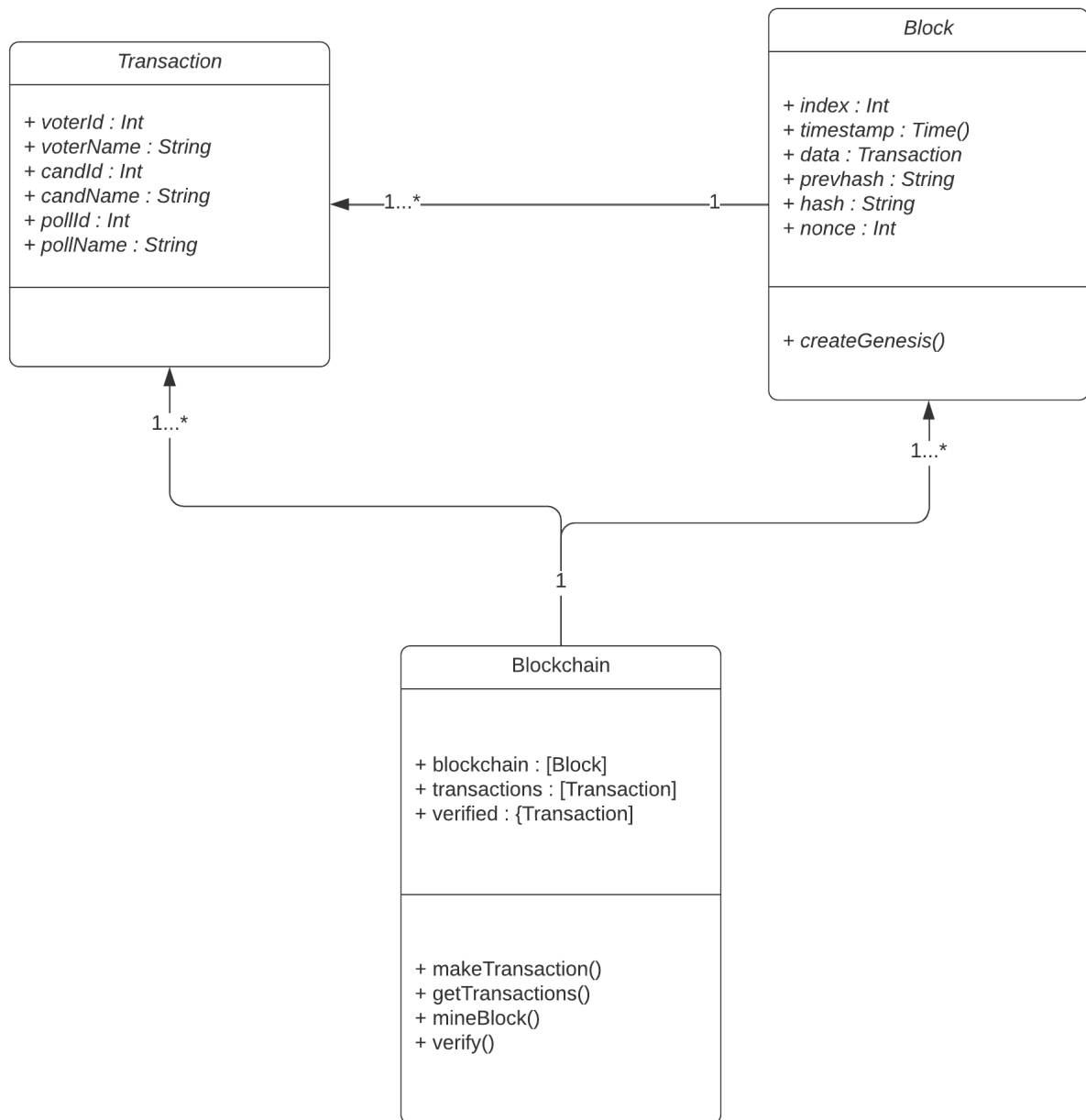
Server -

The endpoints are made using the **flask** framework. The endpoints receive requests sent by the client using **axios**, and respond with a json object. A relational database SQLite is used to store user, candidate and poll data.

Code Functions

- **createGenesis()** : Creates the genesis block of the blockchain.
- **makeTransaction()** : Creates a new transaction (in our case casting a vote) using the given arguments, namely voterId, voterName, candId, candName, pollId and pollName. These arguments specify the id of the voter, the name of the voter, id of the candidate to whom the vote is being cast, name of that candidate, id of the poll in which the vote is being cast and finally the name of the poll respectively.
- **verifyTransaction()** : Takes as an argument the transaction (implemented using a data structure) and uses that to verify the transaction using the zkp implemented.
- **mineBlock()** : Mines a block of the blockchain. Takes as input the current difficulty level and creates a new block, whose POW is set for the given difficulty. This block is then verified for its timestamp and appended accordingly to the blockchain.
- The constructor of the class 'block' takes as argument index, timestamp, data, prevhash, hash, nonce and returns a block with the specified parameters.
- **viewUser()** : Takes as input the user ID and returns the details of the voter's successful transactions in the blockchain.

UML Diagram and Basic Workflow



Database Visualization



The database contains 3 tables, namely, candidates, polls and users -

id	pollId	name	desc
1	1	Ayush Pote	Age - 21 Speciality - Ghot
2	1	Aayush Shah	Age - 20 Speciality - Snake
3	1	Kanishk Yadav	Age - 20 Speciality - Ultra Ghot
4	1	Nayan Sharma	Age - 21 Speciality - Smart :)
5	1	Swapnil Pokale	Age- 21 Speciality - GS

Candidate Table

id	name	password
2019A7PS0114H	Nayan Sharma	NAYAN@123
2019A7PS0112H	Ayush Pote	ayush

User Table

id	name	desc
1	Sample poll	This is a description
2	Another Sample Poll	This is also a description

Polls Table

Screenshots of Working Application



Login and start voting!

Login

Don't have an account? [Register](#)

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Login Page



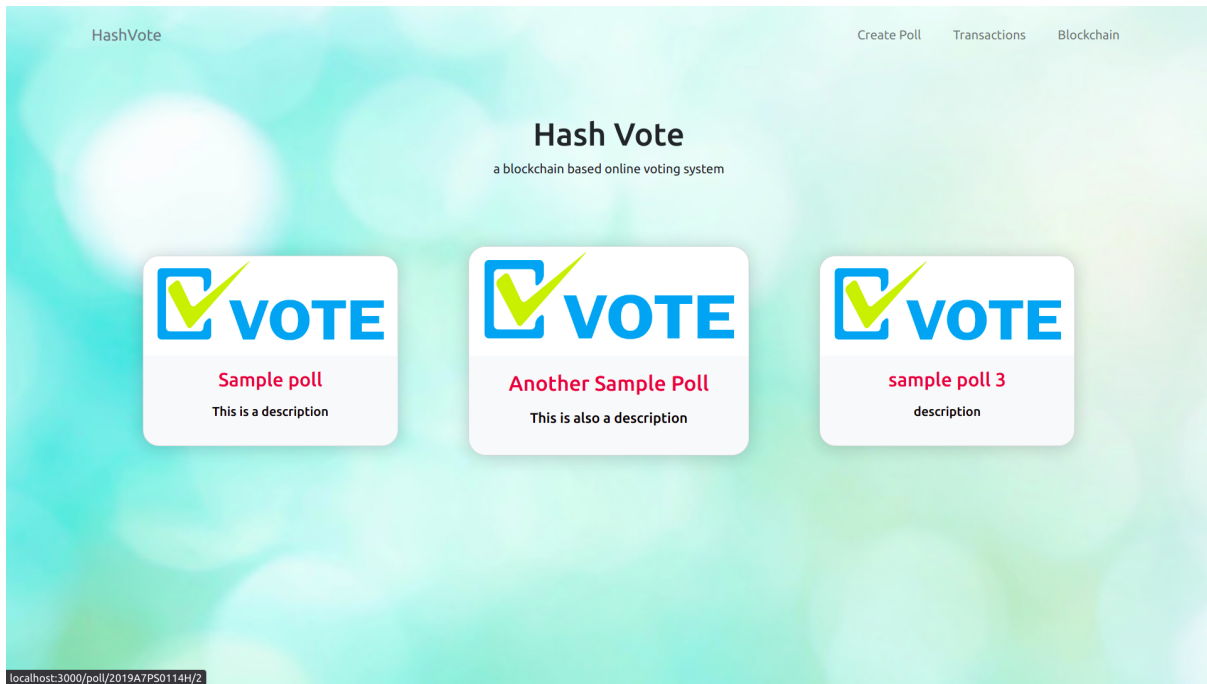
Register and start voting!

Submit

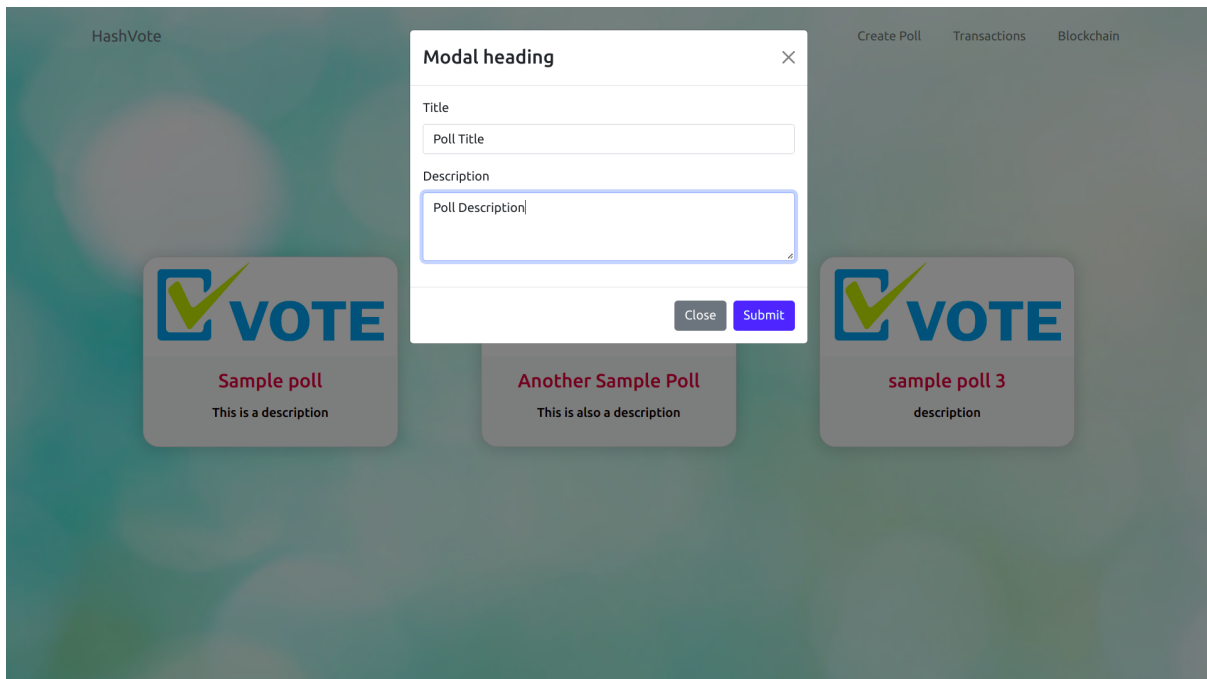
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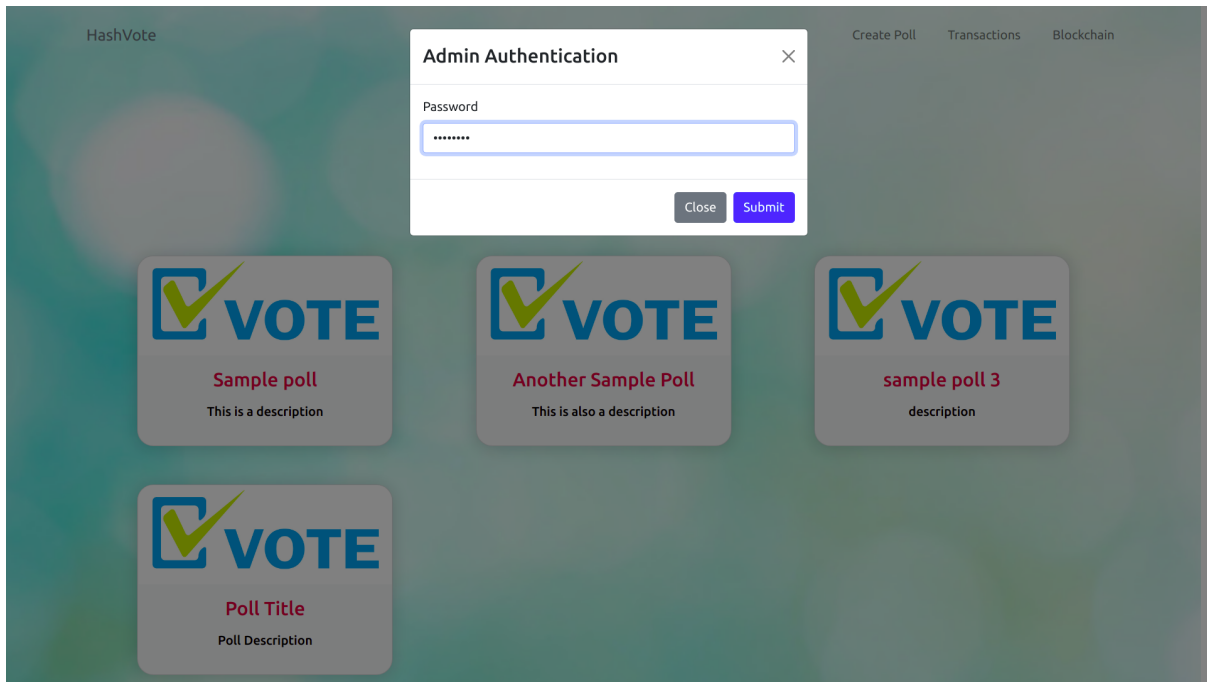
Register Page



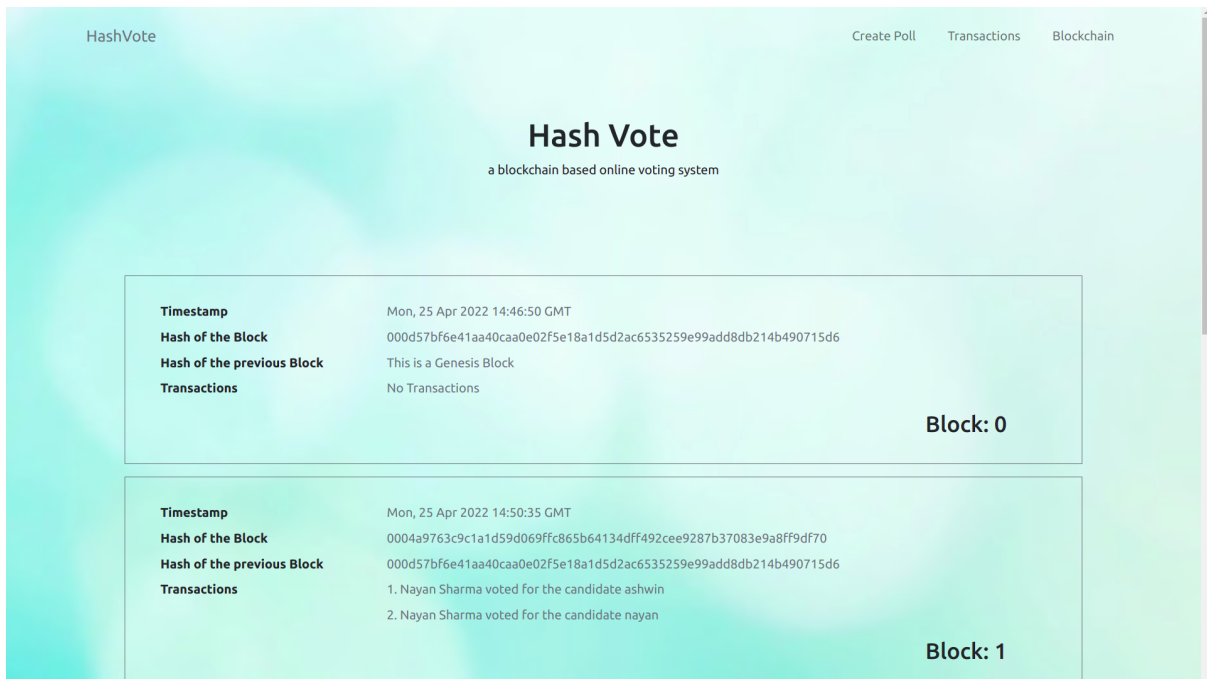
Home Page



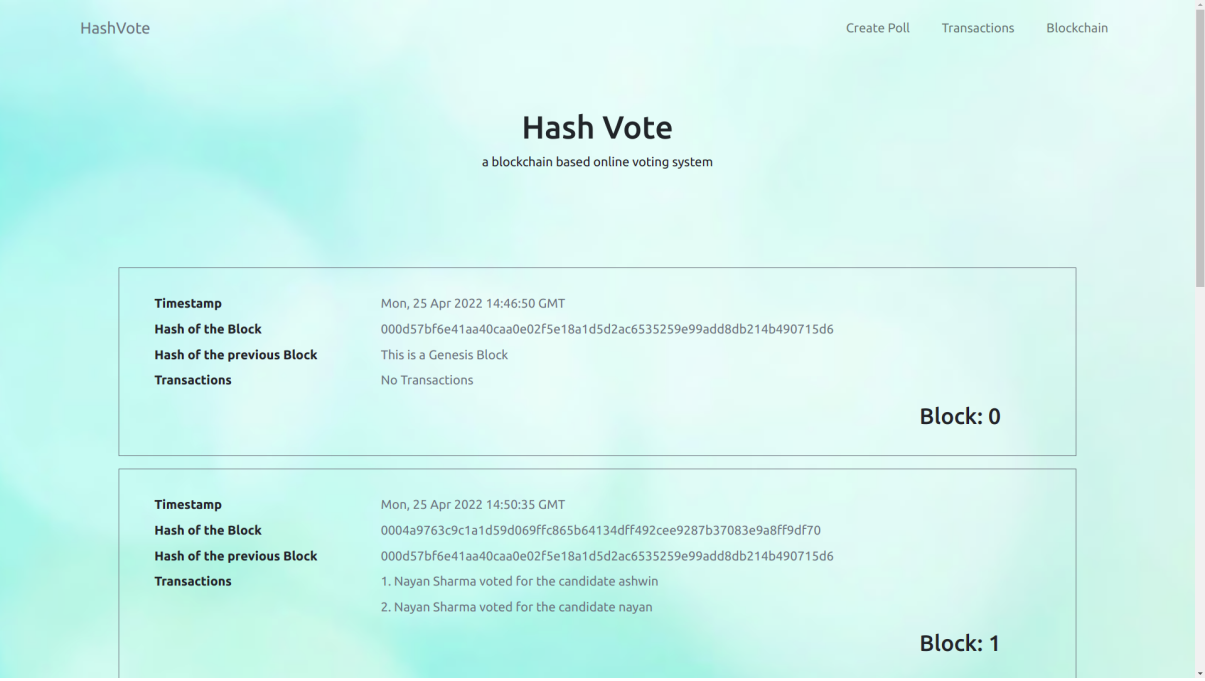
Create Poll component



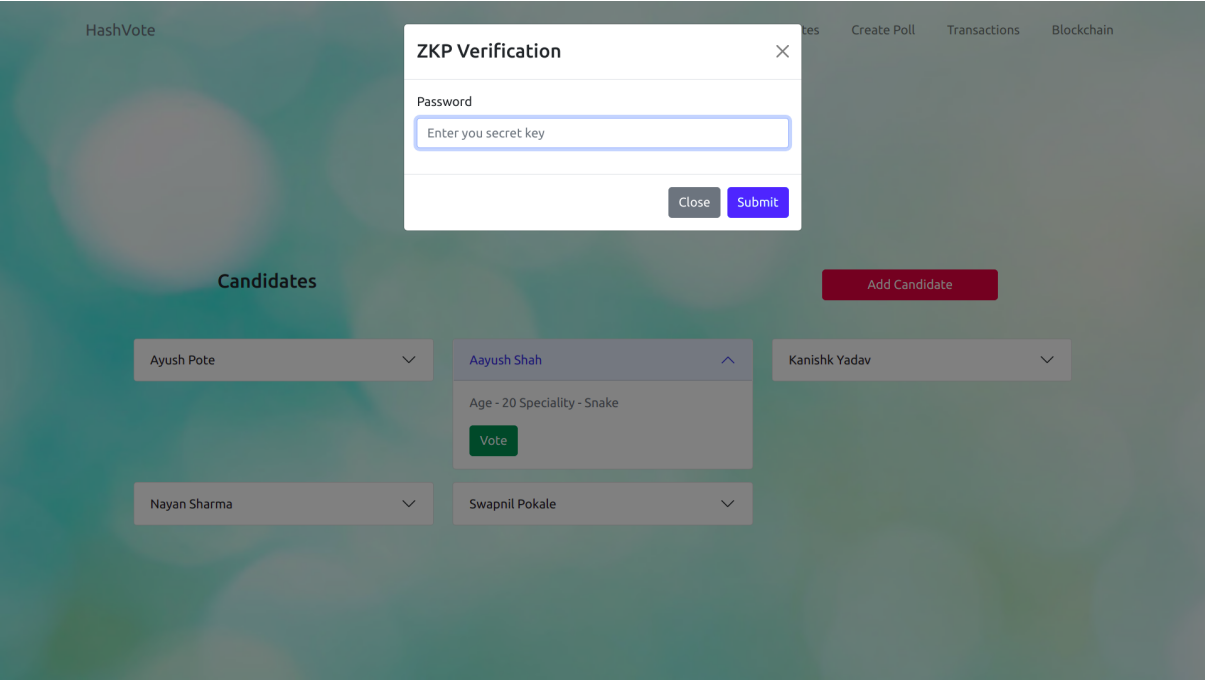
Admin authentication to see the blockchain



User Transactions



Complete Blockchain



ZKP Verification